La Scuola d'Italia / Huson / Physics Nickname:
15 October 2026

2.1 Warm-up: Mental Math (no calculator)

Work each question in your head — no calculator, and no need to show work. Write just the answer, with its units.

Metric prefixes. giga (G) = 1,000,000,000 mega (M) = 1,000,000 kilo (k) = 1000 centi (c) = $\frac{1}{100}$ milli (m) = $\frac{1}{1000}$ micro (μ) = $\frac{1}{1{,}000{,}000}$
A micrometer (µm) is also called a *micron*.

1. Fill in the blanks.

   (a) 2.5 km = _____ m

   (b) 450 cm = _____ m

   (c) 0.8 m = _____ cm

   (d) 35 mm = _____ cm

   (e) 1 m = _____ mm

   (f) 7 mm = _____ m

2. In 1791 the meter was defined so that the North Pole is exactly 10,000 km from the equator. How many meters is that? _____

3. Printer paper is 0.1 mm thick. How many sheets make a 1 cm stack? _____

4. A phone has 256 GB of storage.

   (a) How many megabytes (MB) is that? _____

   (b) A photo takes about 4 MB. About how many photos fit? _____

5. A recent phone's processor runs at 4 GHz: its clock ticks 4 giga-times every second. Write that number of ticks in digits. _____

6. A human red blood cell is about 8 µm across. A phone's processor chip is about 1 cm across.

   (a) How many micrometers are in 1 mm? _____

   (b) How many red blood cells, lined up edge to edge, make 1 mm? _____

   (c) How many micrometers across is the phone chip? _____

7. Write each fraction as a decimal.

   (a) $\frac{1}{4} =$ _____ (c) $\frac{1}{5} =$ _____ (e) $\frac{1}{8} =$ _____ (g) $\frac{1}{3} =$ _____

   (b) $\frac{3}{4} =$ _____ (d) $\frac{1}{10} =$ _____ (f) $\frac{5}{8} =$ _____ (h) $\frac{2}{3} =$ _____

8. Fill in the blanks.

   (a) $\frac{1}{4}$ hour = _____ min

   (b) $\frac{1}{3}$ hour = _____ min

   (c) 0.1 hour = _____ min

   (d) 30 s = _____ min

   (e) 45 min = _____ h (decimal)

   (f) 6 hours = _____ day (fraction)

   (g) 8 hours = _____ day (fraction)

   (h) 1.2 hours = _____ h _____ min

9. Fill in the blanks.

   (a) 1 hour = _____ s (b) 1 year = _____ days (c) 1 year $\approx$ _____ weeks

10. Light travels about 300,000 km every second. The Sun is about 150,000,000 km from the Earth.

    (a) How many seconds does sunlight take to reach us? _____

    (b) Write that time in minutes and seconds. _____ min _____ s

11. Circle the best answer.

    (a) One inch is how many centimeters? (A) 1.6 (B) 2.54 (C) 3.28 (D) 10

    (b) A kilometer is about what fraction of a mile? (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) $\frac{3}{2}$

12. A 5K race is 5 kilometers long. About how many miles is that? _____